Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Florian Klein	Email	Florian.Klein@idmt.fraunhofer.de
Institute Name	Fraunhofer Institute for Digital Media Technology IDMT			
Address	Ehrenbergstrasse 31, 98693 Ilmenau, Germany			

2. Technology Overview

Title of Proposed Technology	Flexible Vibro-Acoustic Sensor Tape for Condition Monitoring			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical · Electronics		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Measuring Instruments	Measurement Sensors and Components	Software	Software solutions
Concept of Proposed Technology	The proposed technology is a flexible, vibro-acoustic sensor tape for condition monitoring, such for industrial machines or structures. Different machine states can be classified on vibrational patterns measured at a given surface as well as the emitted airborne sound. The tape integrates MEMS-based vibration sensors and microphones into a modular array on a flexible substrate. It measures both structure-borne and airborne sound and can be attached to a wide range of three-dimensional machine surfaces without mechanical modification of the equipment. The tape provides a spatially distributed sensing area instead of discrete single sensors. This relaxes requirements on exact sensor placement and enables sound field analysis, source localization and especially AI-based interpretation of state and condition. The concept combines flexible hybrid electronics, dense MEMS sensor integration, and advanced signal processing (including sensor self-localization).			

Rationale for Joint R&D Project	Acoustic process and event monitoring allows early detection of anomalies and failures in industrial environments [1]. Most existing machines lack integrated acoustic sensor systems and cannot be easily retrofitted. At the same time, recent progress in flexible electronics [2] and MEMS devices enables new form factors [3,4], but there is no commercially available solution that combines flexible airborne and structure-borne sensing with array-based signal processing in a flexible, thin and tape-like format. Such a form factor would open totally new opportunities for condition monitoring, for example as retrofit solutions for existing machines or structures or as embedded sensing layer in new machines or structures. Fraunhofer IDMT contributes expertise in acoustic and vibro-acoustic monitoring and array-based signal processing. A Korean SME partner can contribute complementary competences in flexible electronics, sensor integration, and industrialization. A joint R&D project is appropriate to: 1) develop and optimize a manufacturable sensor tape hardware platform, 2) adapt the solution to concrete Korean industrial use cases and environments, 3) prepare scalable production, qualification, and market introduction in Korea and beyond. This joint approach significantly reduces technical uncertainty, enhances the global applicability of the proposed technology by aligning the technology with actual user needs.
Key Words	Condition Monitoring Flexible sensor electronics Vibro-acoustic analysis

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3	End of the Project	Level 6
R&D Objectives	**Overall objective:** Develop and validate a modular acoustic sensor, combining flexible electronics with array-based acoustic sensing and AI-based interpretation of condition and state. **Specific objectives:** 1) Develop a prototype of a flexible sensor array for airborne and structure-borne sound on machine surfaces. 2) Realize a modular design that allows free arrangement of sensors 3) Implement automatic estimation of the 3D positions of individual sensors 4) Integrate a multi-channel audio bus and sensor-near audio preprocessing 5) Enable localization and visualization of airborne and structure-borne acoustic events 6) Validate the complete system in relevant industrial environments under realistic operating conditions.			
R&D Target Specifications	**Target hardware specifications:** ▪ ≥ 16 sensors (microphones and/or accelerometers) per 1000cm² ▪ Modular design (shape, number of sensors) and a minimum bending radius: < 5 cm ▪ Reliable coupling and removability on relevant surfaces ▪ Integrated multi-channel audio bus with at least 16 channels ▪ Operation under typical industrial temperature ranges and vibration levels ▪ Robustness against mechanical bending, fatigue, and moderate contamination **Target Software specifications:** ▪ Synchronized acquisition of airborne and structure-borne signals ▪ Sensor-near preprocessing for feature extraction and data rate reduction ▪ Automatic estimation of sensor positions with accuracy sufficient for sound field reconstruction and event localization ▪ Automatic classification of states and conditions based on multi-point measurements **Validation targets:** ▪ Validation of the system performance on the use-case of a Korean industry partner ▪ Comparison of analysis performance between single-point and multi-point measurements			
R&D Plan	**WP1 Sensor Tape Development and Fabrication:** ▪ Material screening and selection for flexible substrate, adhesive layers, and encapsulation ▪ Flexible circuit and mechanical design including multi-channel audio bus and power distribution ▪ Integration of MEMS microphones and accelerometers using suitable assembly and packaging processes ▪ Fabrication of functional samples and first prototypes; basic electrical, mechanical, and acoustic tests **WP2 Signal Processing and Sensor Self-Localization:** ▪ Development of methods for sensor self-localization on arbitrary 3D surfaces based on acoustic or vibro-acoustic signals. ▪ Design of sound field reconstruction and source localization algorithms for non-uniform arrays ▪ Implementation of sensor-near preprocessing (feature extraction, compression) and compensation of structural vibrations in airborne sound measurements			

	WP3 System Integration and Demonstrator: ▪ Integration of sensor tape hardware (WP1) with signal processing and interfaces (WP2). ▪ Implementation of data transfer to interpretation unit (edge device, PC, cloud) ▪ Development of calibration and basic visualization tools for event localization on the machine surface **WP4 Validation in Industrial Environment:** ▪ Installation and operation of the system on pilot machines in collaboration with the Korean SME and end users ▪ Example analyses (e.g. detection of loosened parts, abnormal friction, process deviations) based on anomaly detection or self-supervised learning depending on available data ▪ Evaluation of installation effort, robustness, and diagnostic performance under real operating conditions. ▪ Refinement of hardware and algorithms where necessary.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute (Fraunhofer IDMT) will lead the core algorithmic research and system-level design of the proposed technology. More specifically this will be: • Development of core algorithms for: sensor self-localization on arbitrary geometries, sound field reconstruction and source localization and event detection as well as classification (e.g. anomalies, impacts, friction events) • Design and implementation of sensor-near preprocessing and data reduction • Development of software tools for analysis and visualization of acoustic data • Laboratory experiments, benchmarking, and support for pilot installations and performance evaluation. In addition, Fraunhofer IDMT will provide research infrastructure, computing resources, and expertise in acoustic signal processing and machine learning, and will be responsible for disseminating scientific outcomes through publications, patents, and technical documentation.

Expected R&R of Korean SME Partner	The Korean SME partner will play a key role in the hardware realization, real-world deployment, field validation, and commercialization-oriented development of the proposed technology. More specifically this will be:
	• Development of the flexible sensor tape hardware, including:
	  ○ design of flexible circuit layouts and mechanical structure,
	  ○ selection and qualification of materials (substrate, adhesive, encapsulation),
	  ○ integration of MEMS sensors, audio bus, and associated electronics on the tape
	• Development of assembly, encapsulation, and reliability concepts (bending, vibration, environmental stress)
	• Fabrication and testing of tape prototypes and iterative design optimization
	• Provision of industrial use cases, machines, and environments for requirements definition and pilot deployment
	• Execution of reliability tests under industrial conditions and feedback to design and algorithms
	• Initiate commercialization activities in Korea and, where appropriate, other Asian markets, including scaling of production and certification

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical outcomes:** The outcome of the project is a prototypical demonstration (TRL 6) of the envisaged sensor tape and corresponding algorithms for detecting the states and conditions of a machine or structure. The benefit of densely measured vibro-acoustic data for estimating these states and conditions will be shown on the example of a specific use-case. **Industrial and economic impact** The solution significantly improves condition monitoring capabilities, supporting predictive maintenance and reducing unplanned downtime in industrial applications. This creates new business opportunities for Korean SMEs in sensor products, data analytics, and monitoring services. Furthermore, it provides a basis for future variants targeting additional sectors such as construction (structural health monitoring) [5] or medical technology (health monitoring) [6], for example. **Scientific and technological impact** The project delivers significant scientific and technological impact by advancing flexible hybrid electronics for vibro-acoustic sensing. It demonstrates dense, self-localizing acoustic arrays operating reliably in industrial environments and contributes new methods for combined structure-borne and airborne vibro-acoustic analysis using AI-based interpretations.

6. References

(1) Becht, P., Deckers, E., Claeys, C., Pluymers, B., Desmet, W., “Loose bolt detection in a complex assembly using a vibro-acoustic sensor array”, Mechanical Systems and Signal Processing, Vol. 130, 2019,pp. 433-451, https://doi.org/10.1016/j.ymssp.2019.05.019

(2) Rim, Y.S., Bae, S.-H., Chen, H., De Marco, N. and Yang, Y., “Recent Progress in Materials and Devices toward Printable and Flexible Sensors”, Adv. Mater., 28: pp. 4415-4440,2016, https://doi.org/10.1002/adma.201505118

(3) Dementyev, A., Kao, H.-L., Paradiso J. A., “SensorTape: Modular and Programmable 3D-Aware Dense Sensor Network on a Tape”, In: Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology (UIST '15). Association for Computing Machinery, New York, NY, USA, 2015, pp.649–658. https://doi.org/10.1145/2807442.2807507

[4] Erbacher, K., Marques, J. A., Krshiwoblozki, M. v., Wu, L., Ngo, H.D., Schneider-Ramelow, M., “Aero Acoustic MEMS Microphone Integration in Ultra-Thin and Flexible Substrate”, IEEE 24th Electronics Packaging Technology Conference (EPTC), pp. 15-18, 2022, DOI: 10.1109/EPTC56328.2022.10013281.

(5) Xinlin P. Qing, Roy Ikegami, Shawn J. Beard, David Zhang, Samik Das, Sourav Banerjee, Fu-Kuo Chang, "Multifunctional sensor network for structural state sensing and structural health monitoring," Proc. SPIE 7647, Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems, 2010, 764711, 2010, https://doi.org/10.1117/12.847180

(6) Zoschke, K., Löher, T., Kallmayer, C., & Jung, E., “Flexible and Stretchable Systems for Healthcare and Mobility. Flexible, Wearable, and Stretchable Electronics”, pp. 269-282, 2020, ISBN: 9780429263941

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization